Mandatory Assignment 3

May 1, 2026

General remarks

- You have to hand in one (!) .qmd (Quarto) script with commented source code in Python or R and one (!) .pdf report in which you describe your methods and results (e.g., with figures, tables, equations, etc.)
- The report (font size 12pt, use the provided template) may not exceed 7 pages including figures, tables, and other appendices
- The source code has to create every figure, table, and number used for your report
- When reporting numbers in your report, please consider a meaningful rounding scheme (there is usually no need to report six digits after the comma)
- Within the report, describe every significant step and make sure that tables and figures are self-explanatory by providing meaningful captions and variable names. The reader will not consult your code — descriptions, captions, and variable names must be self-contained.
- Meaningful code comments explaining the purpose of each step improve both the quality of peer feedback and your final grade
- **Minimum requirement**: the code must run without error or interruption. Ensure the code loads all required packages at the beginning of the script (you can assume that the user has the packages available locally, so do not include `install.packages()` commands). Attempt to solve every task. Document your struggles if you do not finish an exercise. **Hand-ins that do not fulfill this minimum requirement will not receive peer feedback and will not be considered for the exam qualification requirements**
- The assignments can be written individually or by groups of a maximum of three students
- Observe the university's plagiarism rules; for group submissions, note the additional rules for co-written work
- All parts must be answered in English, including comments in the code

The deadline to hand in the files on Peergrade via Absalon is **May 10th, 2026, 8PM**.

Transaction cost robust portfolio backtesting (inspired by Hautsch and Voigt [2019])

Translating return forecasts into tradeable portfolios requires confronting three practical frictions simultaneously: parameter uncertainty in large covariance matrices, model uncertainty in expected return estimates, and the cost of rebalancing. In this assignment, you implement a monthly portfolio backtest that addresses all three.

This assignment connects directly to Mandatory Assignment 2: use the predicted excess returns from your fitted machine learning models in MA 2 as the expected return input $\hat{\mu}_t$. Upload your fitted model objects as part of the submission, as you did for MA 2.

Exercise

Exercises build on each other. The covariance estimators from Exercise 3 and the expected return estimates from Exercise 4 feed directly into the backtesting engine in Exercise 5. Choose and document your estimators clearly, as they determine all downstream results.

0. Go to evaluering.ku.dk and leave your feedback on the course. **Do not move on without filling out the form**

1. *Goal: construct a clean investment universe.* Use the CRSP monthly dataset, starting from January 1980. Retain only stocks with prices above \$2 and a market capitalization above the 20th NYSE size percentile in each month. Report how many stocks N your universe contains on average and briefly characterize the sample.

2. *Goal: establish the theoretical foundation.* Consider the transaction-cost-adjusted certainty-equivalent maximization problem with risk aversion γ:

$$\omega^*_{t+1} := \arg \max_{\omega \in \mathbb{R}^N,\ \iota'\omega=1} \omega'\hat{\mu}_t - \frac{\gamma}{2}\omega'\hat{\Sigma}_t\omega - \lambda(\omega - \omega_{t^+})'\hat{\Sigma}_t(\omega - \omega_{t^+})$$

   in which $\omega_{t^+} := \omega_t \circ (1 + r_t)/\iota'(\omega_t \circ (1 + r_t))$ is the pre-rebalancing weight vector and $\circ$ denotes element-wise multiplication.

   a. Derive the closed-form solution ω^*_{t+1}. Show that it is a convex combination of the unconstrained mean-variance portfolio and ω_{t^+}, and interpret the role of λ.
   b. Discuss whether modeling transaction costs as proportional to portfolio variance is realistic. What does this specification imply about the cost of trading volatile versus stable stocks?
   c. Implement a function `compute_weights(mu, Sigma, gamma, lambda, w_prev)` that returns ω^*_{t+1}. Set $\gamma = 4$ throughout the entire assignment.

3. *Goal: estimate the covariance matrix of returns.* Use a rolling window to produce $\hat{\Sigma}_t$ at each point in time. Suitable options include a factor model (e.g., using Fama-French factors to impose structure on Σ), a DCC-GARCH model, or any well-motivated alternative. Briefly describe its structure and any tuning decisions.

   a. Apply Ledoit-Wolf linear shrinkage on top of your baseline estimator to obtain $\hat{\Sigma}^{LW}_t$.
   b. Compare $\hat{\Sigma}_t$ and $\hat{\Sigma}^{LW}_t$: do they produce meaningfully different (minimum variance) portfolio weights?

4. *Goal: estimate expected returns.*

   a. Use the predicted excess returns $\hat{\mu}_t$ from your MA 2 machine learning models as the primary approach.[1]

5. *Goal: run a portfolio backtest and compare strategies.* Use **January 2010 to December 2024** as the fixed out-of-sample evaluation period. At each month t in this window, construct $\hat{\Sigma}_t$, $\hat{\Sigma}_t^{LW}$, and $\hat{\mu}_t$ using only data available up to t. Set $\lambda = 1000\,\text{bp} = 0.1$.

   Implement and evaluate the following four strategies:

   a. **Naive diversification**: $\omega_t = \frac{1}{N}\iota$, rebalanced monthly
   b. **TC-adjusted MV**: optimal weights from Exercise 2 using $\hat{\Sigma}_t$ and $\hat{\mu}_t$
   c. **TC-adjusted MV (LW)**: same optimization using the Ledoit-Wolf alternative $\hat{\Sigma}_t^{LW}$
   d. **Short-sale constrained MV**: mean-variance portfolio subject to $\omega \geq 0$ in the spirit of Jagannathan and Ma [2003], using $\hat{\Sigma}_t^{LW}$ and $\hat{\mu}_t$

   Report the following out-of-sample statistics for all four strategies in a single summary table:

   - Annualized Sharpe ratio (net of transaction costs)
   - Annualized mean excess return (net of transaction costs)
   - Average monthly turnover (as in equation 46 of Hautsch and Voigt [2019])

   Discuss the results: which strategy performs best after transaction costs? Does the choice of covariance estimator matter? Would you consider this a "true" out-of-sample experiment, and why?

6. Write a 2 - 3 sentence summary of your personal key takeaways from Cilie Feldagers guest lecture (does not count to the page limit).

References

Nikolaus Hautsch and Stefan Voigt. Large-scale portfolio allocation under transaction costs and model uncertainty. *Journal of Econometrics*, 212(1):221–240, 2019. doi: 10.1016/j.jeconom.2019.04.028.

Ravi Jagannathan and Tongshu Ma. Risk reduction in large portfolios: Why imposing the wrong constraints helps. *The Journal of Finance*, 58(4):1651–1683, 2003. doi: 10.1111/1540-6261.00580.

[1] You may instead use moments implied by a suitably chosen multi-factor model or a rolling historical mean. Document your choice.